

Pharmacovigilance Case Assessment

Drug Safety Associate Workup: ADR identification, reporting, causality and severity

Case ref: PV-SIM-2026-0417 | Assessment date: 30 Sep 2026 | Prepared for: B.Pharm exam practice

Case at a glance: Carbamazepine-associated **Stevens-Johnson syndrome** | **Serious** (hospitalisation, medically important) | Causality: **Probable** (WHO-UMC), **Naranjo 7**, **ALDEN 6 (very probable)** | Expected (listed) | Type B (Type IV hypersensitivity) | Severity: Hartwig-Siegel **Level 4b**, CTCAE **Grade 3** | Reporting: PvPI form and CIOMS-I within **15 calendar days (due 15 Oct 2026)**

1. Source case narrative (spontaneous report)

A treating neurologist reported by email on 30 Sep 2026 that a 34-year-old male (initials R.K., 62 kg) developed a severe mucocutaneous reaction after starting carbamazepine for newly diagnosed generalised tonic-clonic epilepsy. Carbamazepine 200 mg orally twice daily was started on 03 Sep 2026 (Day 1) with no titration history. On 19 Sep (Day 17) he developed sore throat, malaise and fever (39 C). On 20 Sep (Day 18) a painful erythematous-to-dusky macular rash appeared on the trunk and spread to the face and limbs. On 21 Sep (Day 19) he developed oral, conjunctival and genital erosions and was admitted. Carbamazepine was stopped on 21 Sep. Paracetamol was first given in hospital on Day 19, after rash onset.

Findings: epidermal detachment about 8% BSA with positive Nikolsky sign; skin biopsy showed full-thickness epidermal necrosis with subepidermal split; ALT 78 U/L (mildly raised); eosinophils normal; Mycoplasma IgM, HSV PCR and HIV negative; urea, glucose and bicarbonate normal. **Treatment:** IV fluids, wound care, topical antiseptics, ophthalmology review, short-course systemic corticosteroid; levetiracetam substituted for seizure control. **Outcome at report:** recovering, re-epithelialisation under way, still hospitalised. No prior drug allergy, no family history of severe drug reactions, no previous carbamazepine exposure. HLA-B*1502 not tested.

Drug	Role	Dose / route	Start	Stop	Indication
Carbamazepine (brand not stated; lot no. XYZ-2417 provided)	Suspect	200 mg BD, oral	03 Sep 2026	21 Sep 2026	Epilepsy (GTCS)
Folic acid	Concomitant	5 mg OD, oral	03 Sep 2026	Ongoing	Supplement
Paracetamol	Concomitant (post-onset)	500 mg SOS, oral	21 Sep 2026	Ongoing	Fever
Levetiracetam	Concomitant (post-onset)	500 mg BD, oral	22 Sep 2026	Ongoing	Seizure control

2. Case intake and validity check (ICSR minimum criteria)

Minimum criterion (ICH E2D)	Present?	Detail
Identifiable patient	Yes	Initials R.K., age 34, male, weight 62 kg
Identifiable reporter	Yes	Treating neurologist (healthcare professional, contact verified)
Suspect drug	Yes	Carbamazepine 200 mg BD
Adverse event	Yes	Mucocutaneous reaction consistent with SJS

Result: valid case. Clock start (Day 0) = 30 Sep 2026, the date the minimum criteria were first received. Source type: spontaneous, HCP-reported, post-marketing, non-interventional.

3. ADR identification and MedDRA coding

Reported term	MedDRA PT (verify in current version)	SOC	Role
Stevens-Johnson syndrome	Stevens-Johnson syndrome	Skin and subcutaneous tissue disorders	Primary event
Fever	Pyrexia	General disorders and administration site conditions	Associated
Mouth erosions	Stomatitis	Gastrointestinal disorders	Associated
Eye involvement	Conjunctivitis	Eye disorders	Associated
Raised liver enzyme	Alanine aminotransferase increased	Investigations	Associated (mild)

- **Definition (WHO):** a noxious and unintended response to a drug at doses normally used in humans. Event onset 17 days after start with a compatible clinical picture and biopsy confirmation = true ADR, not a medication error or misuse.
- **Classification:** Rawlins-Thompson **Type B** (bizarre, not dose-related). Gell and Coombs **Type IV** (delayed, T-cell mediated cytotoxic). Also fits SMQ 'Severe cutaneous adverse reactions'.
- **SJS vs TEN:** detachment under 10% BSA = SJS; 10-30% = SJS/TEN overlap; over 30% = TEN. Here about 8%, so **SJS**.
- **Expectedness:** SJS/TEN is listed in the carbamazepine product information (with HLA-B*1502 warning), so the event is **expected (listed)**.

4. Seriousness assessment (ICH E2A)

Seriousness criterion	Met?	Rationale
Death	No	Patient alive
Life-threatening	No	Not at immediate risk of death at onset (SCORTEN 0-1)
Hospitalisation (initial or prolonged)	Yes	Admitted Day 19 because of the reaction
Persistent or significant disability	No	Not established at this time (follow up ocular sequelae)
Congenital anomaly	No	Not applicable
Other medically important event	Yes	SJS is on the EMA Important Medical Events list

Conclusion: SERIOUS, EXPECTED ADR. SCORTEN (age under 40, no malignancy, HR under 120, BSA under 10%, urea, glucose and bicarbonate normal) = 0 to 1, predicted mortality about 3%.

5. Severity classification

Key exam point: *seriousness* (regulatory outcome-based criteria) is not the same as *severity* (intensity or grade). A mild event can be serious; a severe headache may be non-serious.

Scale	Grade / level	Reason
Hartwig and Siegel (modified)	Level 4b (moderate)	Drug withdrawn, treatment required, and the ADR was the reason for admission. No ICU (would be Level 5), no permanent harm (Level 6), no death (Level 7).
CTCAE v5.0 (SJS)	Grade 3	Skin sloughing under 10% BSA with associated signs (erythema, purpura, epidermal detachment, mucous membrane detachment); hospitalisation indicated.
Clinical (mild / moderate / severe)	Severe	Life-threatening potential, multisystem mucosal involvement, inpatient care.

Hartwig levels: 1-2 mild, 3-4 moderate, 5-7 severe. Where scales disagree, record the reporter's term and your own scale, and state clearly which scale was used.

6. Causality assessment

A. WHO-UMC system

WHO-UMC criterion (Probable / Likely)	Met?
Event with reasonable time relationship to drug intake (onset Day 18 of therapy)	Yes
Unlikely to be attributed to disease or other drugs (Mycoplasma, HSV, HIV negative; paracetamol started after onset; folic acid low risk)	Yes
Response to withdrawal clinically reasonable (progression halted about Day 24, re-epithelialisation)	Yes
Rechallenge not required (and contraindicated ethically)	Yes

WHO-UMC category: PROBABLE / LIKELY. (Not 'Certain' because rechallenge was not done; HLA-B*1502 status is unknown.)

B. Naranjo algorithm

#	Question	Answer	Score
1	Previous conclusive reports on this reaction?	Yes	+1
2	Event appeared after the suspect drug was given?	Yes	+2
3	Improved when drug stopped or antagonist given?	Yes	+1
4	Reappeared on readministration?	Not done	0
5	Alternative causes that could cause the reaction?	No	+2
6	Reaction with placebo?	Not done	0
7	Drug detected in toxic concentration?	Not measured	0
8	More severe with higher dose / less with lower?	Unknown	0
9	Similar reaction to same or similar drug earlier?	No / unknown	0
10	Confirmed by objective evidence?	Yes (biopsy)	+1
	TOTAL		7 = PROBABLE

Naranjo bands: 9 or more definite; 5-8 probable; 1-4 possible; 0 or less doubtful.

C. ALDEN algorithm (specific to SJS/TEN)

ALDEN item	Finding	Score
Delay from start to index day (5-28 days = suggestive)	17 days	+3
Drug present in body on index day	Definitely present	0
Prechallenge / rechallenge	Not done	0
Dechallenge (drug continued without harm = -2)	Drug stopped	0
Notoriety of drug (strongly associated)	Carbamazepine	+3
Other cause (alternative drug with score above 3)	None; other drugs started after onset or low risk	0
TOTAL		6 = VERY PROBABLE

ALDEN bands: under 0 very unlikely; 0-1 unlikely; 2-3 possible; 4-5 probable; 6 or more very probable.

D. Causality of each drug

Drug	Company causality	Rationale
Carbamazepine	Related (probable)	Time course, known high-risk drug, dechallenge, biopsy, other causes excluded
Folic acid	Unlikely	No known SJS association; weak temporal reasoning
Paracetamol	Unlikely	Started after reaction onset (cannot be the cause)
Levetiracetam	Not related	Started after onset

Reporter causality: 'related'. Company causality: 'related'. Both are recorded on the report; when they differ, both are recorded, never overwritten.

7. Reporting pathway and timelines

Step	Action	Owner / timeline
1	Log receipt, assign case number, check validity, check duplicates	Drug Safety Associate, Day 0 (30 Sep 2026)
2	Code events (MedDRA), drugs (WHO-DD), assess seriousness, expectedness, causality	Drug Safety Associate, Day 1-3
3	Medical review and sign-off of causality and narrative	Safety physician, Day 3-5
4	Prepare CIOMS-I and E2B(R3) ICSR; India: PvPI Suspected ADR Reporting Form	Case processor, Day 5-10
5	Submit: serious case to regulator within 15 calendar days (due 15 Oct 2026) ; PvPI / NCC-PvPI (IPC, Ghaziabad) and, where applicable, global database via VigiFlow to WHO-UMC	Submission team, by Day 15
6	Send follow-up queries; submit follow-up ICSR with new information	Ongoing (target 2 attempts)
7	Include case in PSUR/PBRER and signal review; consider label or RMP update	Periodic / signal management

- Timeline note: a fatal or life-threatening case needs a 7-day initial report and 8-day follow-up. This case is serious but neither fatal nor life-threatening, hence 15 days. In the US, serious and *unexpected* cases are 15-day alerts and serious *expected* ones go in periodic reports; many other regions and PvPI practice expect all serious cases quickly. Always apply the local regulation and company SOP.
- **CIOMS-I mapping:** I = reaction information (dates, outcome); II = suspect drug(s) (dose, route, dates, indication, dechallenge); III = concomitant drugs and history; IV = reporter / manufacturer details.

8. Follow-up questions to the reporter

- HLA-B*1502 (and HLA-A*3101) genotype result, if available; ethnicity and family history.
- Complete lab dates and values (LFT trend, renal function, CBC); ophthalmology findings; total BSA over time.
- Exact steroid agent, dose and duration; ICU need; final outcome and discharge date; any sequelae (ocular, skin pigmentation).
- Brand and manufacturer, batch number, expiry date (rule out product quality issue), any OTC or herbal products used.
- Confirm no earlier exposure to aromatic anticonvulsants (phenytoin, phenobarbital, lamotrigine, oxcarbazepine) and no cross-reactive prior reaction.

9. Preventability, risk minimisation and counselling

- **Preventability (Schumock-Thornton):** potentially preventable. HLA-B*1502 screening is recommended before starting carbamazepine in patients of Asian ancestry, and warning signs (fever, sore throat, rash, mouth sores) should be counselled.
- **Management principles:** stop the culprit immediately, supportive burn-unit style care, avoid all cross-reactive aromatic anticonvulsants, document as an allergy/ADR alert.
- **Patient advice:** permanent avoidance of carbamazepine; carry an ADR card; report new rash and fever early. Consider genetic counselling for first-degree relatives.
- **Signal view:** a known risk with an established association; no new signal from a single case, but include in aggregate review. Risk minimisation: label warning already present; consider reinforcing HLA testing message through a Dear Healthcare Professional communication if reports cluster.

10. Company narrative (regulatory-style)

This spontaneous, serious, expected case was reported by a neurologist and concerns a 34-year-old male who started carbamazepine 200 mg BD on 03 Sep 2026 for epilepsy. After 17 days he developed fever and sore throat followed by a mucocutaneous eruption with oral, ocular and genital erosions and about 8% BSA epidermal detachment. Skin biopsy confirmed full-thickness epidermal necrosis consistent with Stevens-Johnson syndrome. Carbamazepine was discontinued on 21 Sep 2026 and the patient was hospitalised; supportive care, corticosteroids and levetiracetam were given, and the patient was recovering at the time of reporting. Concomitant paracetamol and levetiracetam were started after onset. Infectious causes were negative. HLA-B*1502 was not tested. The reporter considered the event related to carbamazepine; the company assessment is **probable (WHO-UMC), Naranjo 7, ALDEN 6**. Follow-up requested for genotype, final outcome and sequelae.

11. Exam-ready answer structure (use for any case)

Step	What to write	Marks tip
1	Validity: 4 minimum criteria	Name all four
2	ADR identification, type (A/B), coding (MedDRA)	Give PT and SOC
3	Seriousness (ICH E2A) then expectedness	State the criteria met
4	Severity (Hartwig-Siegel / CTCAE), keep separate from seriousness	Say 'not the same'
5	Causality: WHO-UMC and Naranjo (table), ALDEN for SJS/TEN	Show score workings
6	Preventability, management, dechallenge outcome	Mention Schumock-Thornton
7	Reporting: form, agency, timeline, follow-up	PvPI, VigiFlow, 15 / 7 days
8	Narrative and conclusion in 4-5 lines	Include dates in order

Disclaimer: this is a fictional teaching case built for exam practice. Regulatory timelines and coding versions vary by jurisdiction and company SOP; confirm against current PvPI, ICH and CDSCO documents before applying in practice.